

Cluster Overlap Rate in Dirichlet Mixtures

Mohammed Ouali and Bautista Rodriguez
Department of Computer Science, Adrian College
Adrian, MI, 49221, United States
{mouali, brobledo26}@adrian.edu

Abstract—This paper proposes an analytical framework for characterizing overlap between two Dirichlet distributions. Given two Dirichlet probability density functions defined on the same probability simplex, we study the equality set where the two densities take the same value. This set defines an implicit constraint inside the simplex. We then define the maximum-density intersection point as the point on this equality set where the common density is maximized. This point represents the most probable configuration equally supported by both distributions. We derive a constrained optimization formulation, obtain necessary optimality conditions through Lagrange multipliers, and show that the resulting problem can be reduced to solving a one-dimensional nonlinear equation. The associated maximum intersection density is proposed as a local indicator of cluster ambiguity. Experiments on synthetic Dirichlet mixtures are designed to evaluate the relationship between this measure and clustering difficulty.

Index Terms—Dirichlet distribution, compositional data, probability simplex, overlap rate, mixture models, clustering, constrained optimization.

I. INTRODUCTION

Clustering aims at partitioning data into groups of similar observations, typically under implicit or explicit assumptions about the underlying data distribution. A large class of widely used clustering algorithms, including k -means [1] and Gaussian Mixture Models (GMMs) [2], relies on the assumption that data are generated from mixtures of Gaussian distributions [3], [4], [5], [6], [7]. While Fuzzy C-Means (FCM) [8] does not explicitly assume an underlying probabilistic model, its reliance on Euclidean distance to cluster centroids implicitly favors compact and approximately spherical cluster structures, similar to Gaussian-based methods as argued in [3]. As a result, its performance may degrade when the data exhibit non-Euclidean geometry or compositional constraints, where Dirichlet mixture models provide a more appropriate representation. While this assumption is mathematically convenient and often effective in Euclidean spaces, it can be restrictive when dealing with compositional data constrained to the probability simplex or data exhibiting skewness, sparsity, or boundary effects. In contrast, mixtures of Dirichlet distributions provide a more flexible modeling framework for such data, as they naturally enforce positivity and normalization constraints while accommodating a wide range of distributional shapes. This flexibility makes Dirichlet mixtures particularly well-suited for capturing the variability observed in real-world compositional datasets.

Dirichlet distributions play a central role in modeling compositional data, where observations are constrained to lie on

the probability simplex. Such data arise naturally in a wide range of applications, including mixture modeling, image and signal processing, where normalized feature representations such as color histograms and satellite-image histograms are commonly used. In these settings, each data point represents proportions that sum to one, making the Dirichlet distribution a natural and flexible modeling choice.

In clustering and mixture modeling, the degree of overlap rate between clusters is a key factor influencing separability, classification accuracy, and the stability of learning algorithms. When two clusters exhibit significant overlap rate, distinguishing between them becomes inherently difficult, leading to ambiguity in clustering assignments and degradation in performance. As a result, quantifying overlap rate between distributions is of fundamental importance for understanding and evaluating clustering behavior.

Existing approaches to comparing probability distributions rely primarily on global divergence or similarity measures, such as Kullback–Leibler divergence, Jensen–Shannon divergence, and Bhattacharyya distance. While these measures provide useful summaries of overall dissimilarity, they do not explicitly characterize the regions of the domain where two distributions assign equal probability density, nor do they identify the most probable points of ambiguity between them.

In this paper, we introduce an analytical framework for characterizing overlap between two Dirichlet distributions defined on the same simplex. Rather than relying solely on global measures, we focus on the *equality set* formed by the points where the two Dirichlet densities coincide. This set defines an implicit nonlinear constraint within the simplex. We then formulate the problem of identifying the most probable ambiguous configuration as a constrained optimization problem over this equality set.

Specifically, we define the *maximum-density intersection point* as the point on the equality set at which the common density is maximized. This point represents the most likely configuration that is equally supported by both distributions and thus serves as a local indicator of cluster ambiguity. The resulting optimization problem is solved analytically using Lagrange multipliers, leading to a set of necessary optimality conditions. We show that the problem can be reduced to solving a one-dimensional nonlinear equation in a scalar parameter, enabling efficient computation.

Our work extends prior research in the literature on overlap rate estimation in Gaussian mixtures [5], [4] to the Dirichlet case, where the simplex constraint introduces additional

analytical challenges. Unlike Gaussian distributions defined on unconstrained Euclidean spaces, Dirichlet distributions are defined on a bounded domain with normalization constraints, requiring careful treatment of feasibility and optimality conditions.

The proposed framework provides both a structural characterization of the equality set and a computable quantity capturing local overlap. This quantity can be used to analyze ambiguity in Dirichlet mixture models and to study its relationship with clustering performance. Experimental results on synthetic data illustrate how the proposed measure correlates with clustering difficulty.

The remainder of the paper is organized as follows. Section II presents the analytical characterization of the equality set and derives a parametric representation of the curve of intersection. Section III determines the maximum-density intersection point. Section IV introduces the proposed measure of overlap between two Dirichlet distributions and derives its expression. Section V reports and discusses the experimental results. Finally, Section VI concludes the paper and outlines directions for future work.

II. ANALYTICAL CHARACTERIZATION OF EQUALITY SETS

Let us consider a Dirichlet distribution with parameter vector $\alpha = (\alpha_1, \alpha_2, \alpha_3)$, the density is

$$f_\alpha(x_1, x_2, x_3) = \frac{\Gamma(\alpha_1 + \alpha_2 + \alpha_3)}{\Gamma(\alpha_1)\Gamma(\alpha_2)\Gamma(\alpha_3)} x_1^{\alpha_1-1} x_2^{\alpha_2-1} x_3^{\alpha_3-1},$$

with $x_1 > 0, x_2 > 0, x_3 > 0$, and $x_1 + x_2 + x_3 = 1$.

Definition 1 (Maximum-density intersection point). *We define the maximum-density intersection point between two distributions $f_\alpha(x)$ and $f_\eta(x)$ as*

$$x^* = \arg \max_{x \in \mathcal{I}_{\alpha, \eta}} f_\alpha(x). \quad (1)$$

Since $f_\alpha(x) = f_\eta(x)$ for every $x \in \mathcal{I}_{\alpha, \eta}$, maximizing f_α over the equality set is equivalent to maximizing f_η over the same set.

The equal density set is characterized by

$$h(x) = \log C(\alpha) - \log C(\eta) + \sum_{i=1}^K (\alpha_i - \eta_i) \log x_i = 0. \quad (2)$$

The optimization problem can be written in log-density form

$$\max_x \ln f_\alpha(x) \quad (3)$$

$$\text{subject to } h(x) = 0, \quad (4)$$

$$\sum_{i=1}^K x_i = 1, \quad (5)$$

$$x_i > 0, \quad i = 1, \dots, K. \quad (6)$$

Using $x_3 = 1 - x_1 - x_2$, the log-density of f_α is $\ell_\alpha(x_1, x_2) = \ln f_\alpha(x_1, x_2, 1 - x_1 - x_2)$. Up to an additive constant, this is $\ell_\alpha(x_1, x_2) = (\alpha_1 - 1) \ln x_1 + (\alpha_2 - 1) \ln x_2 + (\alpha_3 - 1) \ln(1 - x_1 - x_2)$.

The intersection condition $f_\alpha = f_\eta$ can be written as $h(x_1, x_2) = 0$, where $h(x_1, x_2) = C + (\alpha_1 - \eta_1) \ln x_1 + (\alpha_2 - \eta_2) \ln x_2 + (\alpha_3 - \eta_3) \ln(1 - x_1 - x_2)$, and

$$C = \ln \left(\frac{\Gamma(\alpha_1 + \alpha_2 + \alpha_3)}{\Gamma(\alpha_1)\Gamma(\alpha_2)\Gamma(\alpha_3)} \right) - \ln \left(\frac{\Gamma(\eta_1 + \eta_2 + \eta_3)}{\Gamma(\eta_1)\Gamma(\eta_2)\Gamma(\eta_3)} \right).$$

So the problem is to maximize $\ell_\alpha(x_1, x_2)$ subject to $h(x_1, x_2) = 0$.

The full Lagrangian for this problem is

$$\mathcal{L}(x_1, x_2, x_3, \lambda, \mu) = \sum_{i=1}^3 (\alpha_i - 1) \ln x_i - \lambda \left(\sum_{i=1}^3 (\alpha_i - \eta_i) \ln x_i + C \right) - \mu \left(\sum_{i=1}^3 x_i - 1 \right).$$

Differentiating with respect to x_i gives

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\alpha_i - 1}{x_i} - \lambda \frac{\alpha_i - \eta_i}{x_i} - \mu = 0.$$

Multiplying by x_i and rearranging yields $\mu x_i = (\alpha_i - 1) - \lambda(\alpha_i - \eta_i)$ and by letting $b_i(\lambda) = (1 - \lambda)(\alpha_i - 1) + \lambda(\eta_i - 1)$, we finally have

$$\mu x_i = b_i(\lambda), \quad \text{so} \quad x_i = \frac{b_i(\lambda)}{\mu}.$$

Using the simplex condition $\sum_i x_i = 1$,

$$\sum_{i=1}^3 x_i = \frac{1}{\mu} \sum_{i=1}^3 b_i(\lambda) = 1,$$

hence

$$\mu = \sum_{j=1}^3 b_j(\lambda).$$

Therefore,

$$x_i(\lambda) = \frac{b_i(\lambda)}{\sum_{j=1}^3 b_j(\lambda)}. \quad (7)$$

III. MAXIMUM-DENSITY INTERSECTION POINT

The highest point on the intersection curve is the point where f_α is maximal subject to $f_\alpha = f_\eta$. By substituting 7 into $h(x_1, x_2) = 0$, we obtain a single equation in λ :

$$C + (\alpha_1 - \eta_1) \ln x_1(\lambda) + (\alpha_2 - \eta_2) \ln x_2(\lambda) + (\alpha_3 - \eta_3) \ln x_3(\lambda) = 0.$$

Solving the above equation for λ and substituting in 7, we get the localization of maximum density intersection point. Figure III shows two Dirichlet distributions shown by their densities, their equal density set (white curve), and maximum-density intersection point (point in red color).

IV. MAXIMUM-DENSITY INTERSECTION AS AN OVERLAP INDICATOR

Once x^* has been found, we define

$$O_{\max}(\alpha, \eta) = f_\alpha(x^*) = f_\eta(x^*). \quad (8)$$

The corresponding log-form is

$$\ln O_{\max}(\alpha, \eta) = \ln f_\alpha(x^*). \quad (9)$$

A large value of $O_{\max}$ indicates that the two distributions intersect in a high-density region, suggesting strong local

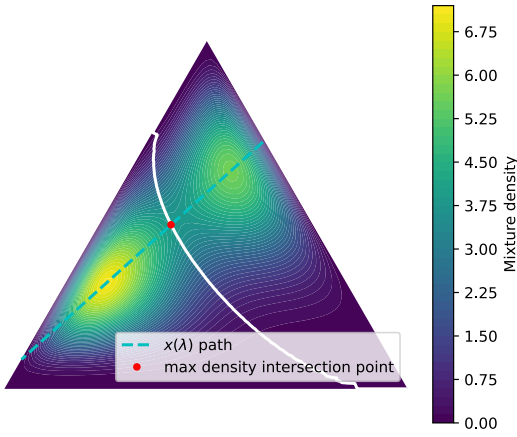


Fig. 1. Two overlapping distributions with their equal density intersection points set and the maximum density intersection point.

ambiguity. A small value indicates that the equality occurs primarily in lower-density regions.

Because raw density values depend on dimension and concentration, a normalized value may also be considered:

$$\tilde{O}_{\max} = \frac{f_{\alpha}(x^*)}{\min\{f_{\alpha}(m_{\alpha}), f_{\eta}(m_{\eta})\}}, \quad (10)$$

where m_{α} and m_{η} are the modes of the two Dirichlet distributions, when they exist.

V. EXPERIMENTAL RESULTS

For $K = 3$, the simplex can be represented as a triangle. We generate pairs of Dirichlet distributions and plot:

- density contours of $f_{\alpha}(x)$,
- density contours of $f_{\eta}(x)$,
- the equality curve $f_{\alpha}(x) = f_{\eta}(x)$,
- the maximum-density intersection point x^* .

Figure 2 shows examples of synthetic data generated according to distributions mixtures with varying parameter settings. Each mixture contains two distributions and each distribution has 2000 points. Figure 2-Top shows two distributions with parameters $\alpha = (8, 2, 2)$ and $\eta = (2, 8, 2)$; Figure 2-Middle shows two distributions with parameters $\alpha = (6, 3, 2)$ and $\eta = (3, 6, 2)$; Figure 2-Bottom shows two distributions with parameters $\alpha = (5, 4, 2)$ and $\eta = (4, 5, 2)$. These cases represent decreasing cluster separation and increasing overlap.

We construct a sequence of Dirichlet pairs with gradually increasing similarity. For each pair, we compute $\tilde{O}_{\max}$ and compare it with clustering performance on samples generated from a two-cluster Dirichlet mixture. The overlap measures, $\tilde{O}_{\max}$, in Fig. 2 are 0.0889 (for Top row mixture), 0.528 (for Middle row mixture), and 0.850.

For each parameter pair, we generated 2000 samples from the mixture. The true cluster labels are retained. Clustering algorithms are then applied, and the predicted labels are compared with the true labels using accuracy, precision, and the Adjusted Rand Index (ARI).

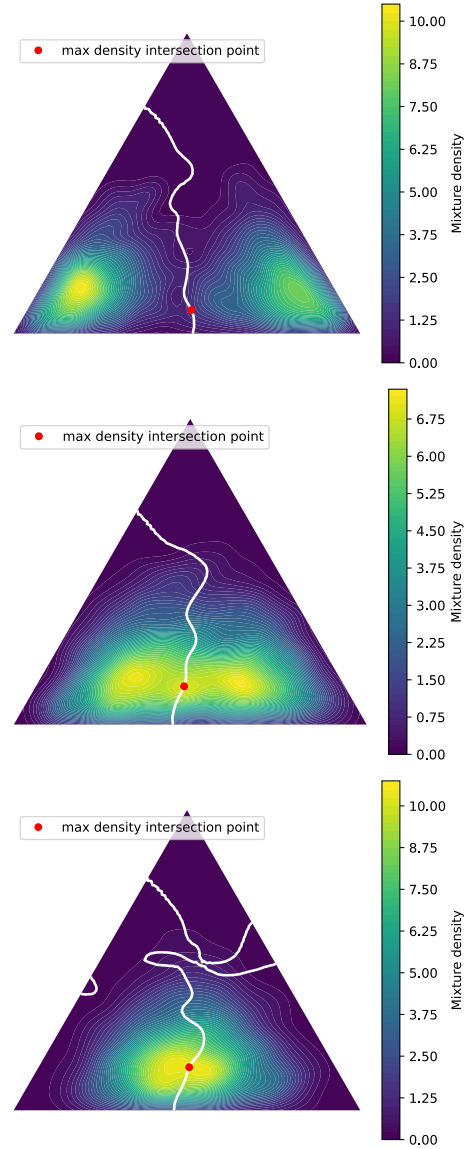


Fig. 2. Generated scattered points and density plots. Top: distributions $\alpha[8,2,2]$ and $\eta[2,8,2]$. Middle: distributions $\alpha[6,3,2]$ and $\eta[3,6,2]$. Bottom: distributions $\alpha[5,4,2]$ and $\eta[4,5,2]$.

Candidate clustering methods include K -means on simplex coordinates, Fuzzy C-means, Gaussian Mixture Model.

Tables I–III report the clustering performance of K-Means, Fuzzy C-Means (FCM), and Gaussian Mixture Models (GMMs) on three synthetic datasets generated from Dirichlet mixtures with increasing levels of overlap, as quantified by the proposed measure $\tilde{O}_{\max}$. In the first scenario, corresponding to a low overlap level ($\tilde{O}_{\max} = 0.0889$), all three methods achieve high performance, with accuracy and precision close to 0.98 and Adjusted Rand Index (ARI) values above 0.91. This indicates that when the clusters are well separated, classical clustering algorithms are able to correctly identify the underlying structure, regardless of the mismatch between the data distribution and the Gaussian assumption. As the overlap

increases to a moderate level ($\tilde{O}_{\max} = 0.528$), a noticeable degradation in performance is observed across all methods. K-Means and FCM maintain comparable performance, with accuracy around 0.86 and ARI near 0.51, while GMM exhibits a more significant drop, particularly in ARI (0.292). This suggests that the Gaussian assumption underlying GMM becomes less suitable as the true data distribution deviates from Gaussianity and overlap increases. In contrast, distance-based methods such as K-Means and FCM appear more robust in this intermediate regime. In the high-overlap scenario ($\tilde{O}_{\max} = 0.850$), clustering performance deteriorates substantially for all methods. Accuracy drops to approximately 0.63 for K-Means and FCM, and to nearly random performance (0.51) for GMM. The ARI values approach zero, indicating that the clustering assignments are nearly uncorrelated with the true labels. This behavior reflects the inherent difficulty of the problem when clusters strongly overlap, where even the most probable configurations become highly ambiguous.

Overall, the results demonstrate a clear relationship between the proposed overlap measure $\tilde{O}_{\max}$ and clustering performance. As $\tilde{O}_{\max}$ increases, all performance metrics consistently degrade, confirming that the measure effectively captures the level of ambiguity between clusters. Furthermore, the results highlight the limitations of Gaussian-based modeling in compositional settings and support the need for distribution-aware approaches, such as those based on Dirichlet mixtures. These results suggest a monotonic relationship between the proposed overlap measure and clustering performance, indicating that $\tilde{O}_{\max}$ can serve as a predictive indicator of clustering difficulty.

TABLE I
CLUSTERING PERFORMANCE ON A DIRICHLET MIXTURE OF TWO DISTRIBUTIONS WITH $\alpha = [8, 2, 2]$ AND $\eta = [2, 8, 2]$. THE ESTIMATED OVERLAP MEASURE BETWEEN THE CLUSTERS IS $\tilde{O}_{\max} = 0.0889$.

Method	Accuracy	Precision	ARI
K-Means	0.984	0.984	0.935
Fuzzy C-Means	0.984	0.984	0.9370
Gaussian Mixture Model	0.979	0.979	0.918

TABLE II
CLUSTERING PERFORMANCE ON A DIRICHLET MIXTURE OF TWO DISTRIBUTIONS WITH $\alpha = [6, 3, 2]$ AND $\eta = [3, 6, 2]$. THE ESTIMATED OVERLAP MEASURE IS $\tilde{O}_{\max} = 0.528$.

Method	Accuracy	Precision	ARI
K-Means	0.857	0.858	0.509
Fuzzy C-Means	0.858	0.858	0.510
Gaussian Mixture Model	0.771	0.804	0.292

VI. CONCLUSION

This paper introduced an analytical framework for characterizing overlap between Dirichlet distributions. The equality set between two Dirichlet probability density functions was expressed as a logarithmic constraint on the simplex, and

TABLE III
CLUSTERING PERFORMANCE ON A DIRICHLET MIXTURE OF TWO DISTRIBUTIONS WITH $\alpha = [5, 4, 2]$ AND $\eta = [4, 5, 2]$. THE ESTIMATED OVERLAP MEASURE IS $\tilde{O}_{\max} = 0.850$.

Method	Accuracy	Precision	ARI
K-Means	0.635	0.635	0.0724
Fuzzy C-Means	0.636	0.858	0.0730
Gaussian Mixture Model	0.512	0.512	0.00006

the corresponding maximum-density intersection point was formulated as a constrained optimization problem. By applying Lagrange multipliers, the problem was reduced to solving a scalar nonlinear equation, enabling efficient computation.

The resulting maximum-density intersection provides a local and interpretable indicator of cluster ambiguity, complementing traditional global divergence measures. Experimental results demonstrated a clear relationship between the proposed measure and clustering performance, highlighting its relevance for analyzing the difficulty of clustering tasks in Dirichlet mixture models.

Although the proposed overlap measure is applicable to Dirichlet mixture models and compositional data represented on the simplex, its effectiveness depends on the suitability of the Dirichlet assumption for modeling the underlying data distribution. Applications involving non-compositional data or distributions that significantly depart from Dirichlet behavior may require alternative overlap formulations.

Future work will investigate extensions to higher-dimensional settings, multi-component mixtures, and the integration of the proposed measure into clustering algorithms for improved model selection and performance evaluation. Additional research will explore overlap measures for generalized Dirichlet distributions and other mixture models to broaden the applicability of the proposed framework.

REFERENCES

- [1] D. Aloise, A. Deshpande, P. Hansen, and P. Popat, "Np-hardness of euclidean sum-of-squares clustering," *Machine Learning*, vol. 75, no. 2, pp. 245–248, 2009.
- [2] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.
- [3] R. Gharbaoui, M. Ouali, and N. Benamrane, "Overlap estimation to hierarchical clustering," *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, vol. 34, no. 02, pp. 143–168, 2026.
- [4] —, "Generation of Gaussian sets for clustering methods assessment," *Data & Knowledge Engineering*, vol. 131–132, p. 101876, 2021.
- [5] —, "Estimating and controlling overlap in Gaussian mixtures for clustering methods evaluation," *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, vol. 28, no. 2, pp. 183–211, 2020.
- [6] —, "A new similarity measure and hierarchical clustering approach to color image segmentation," in *2023 IEEE 24th International Conference on Information Reuse and Integration for Data Science (IRI)*. Bellevue, WA, USA: IEEE, 2023, pp. 34–39.
- [7] M. Ouali, R. Gharbaoui, and N. Benamrane, "Fusion-based clustering with interaction rate for imbalanced and chained structures," in *2025 IEEE 26th International Conference on Information Reuse and Integration for Data Science (IRI)*. San Jose, CA, USA: IEEE, 2025.
- [8] H. Zhang and S. Huang, "Improved fuzzy C-means clustering algorithm based on fuzzy particle swarm optimization for solving data clustering problems," *Mathematics and Computers in Simulation*, vol. 233, pp. 311–329, 2025.